

DELAWARE BAY BUOY 44009, USA

WMO: 992400

Lat: 38.457N

Lon: 74.702W

Elev: 0

StdP: 14.70

Time Zone: -5.00 (W05)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	20.7	24.6	5.9	7.4	23.1	9.5	8.9	25.7	38.2	32.0	36.2	33.6	24.8	310	0.606

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.7	80.1	76.2	79.0	75.6	78.0	74.9	78.1	78.8	77.3	78.1	76.4	77.5	10.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.8	144.5	78.4	77.0	140.5	77.8	76.0	135.7	77.2	41.5	78.8	40.6	78.2	39.8	77.7	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
34.8	31.9	29.3	17.7	83.3	4.5	1.9	14.4	84.6	11.7	85.7	9.2	86.8	5.9	88.1
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.4	38.7	38.4	42.5	49.9	58.3	68.4	73.9	74.4	70.3	62.0	53.2	45.2
	DBStd	13.97	7.74	6.22	5.11	4.33	4.78	4.46	2.79	2.86	3.68	5.24	6.25	7.23
	HDD50	1193	353	326	235	54	2	0	0	0	0	1	41	181
	HDD65	4035	815	746	697	453	212	19	0	0	5	119	354	615
	CDD50	3515	3	0	3	50	260	551	742	755	610	373	137	31
	CDD65	885	0	0	0	6	120	277	291	165	26	0	0	0
	CDH74	2275	0	0	0	0	0	102	885	1074	210	4	0	0
	CDH80	32	0	0	0	0	0	2	14	16	0	0	0	0
	WSAvg	14.3	17.6	16.3	15.5	14.3	12.1	11.3	10.7	11.3	13.5	15.5	16.3	17.5
	PrecAvg	40.8	3.0	2.8	3.9	3.1	2.9	3.1	3.7	3.9	3.8	3.4	3.2	3.8
(15) Precipitation	PrecMax	59.6	6.3	5.8	7.6	5.9	5.6	6.9	7.1	10.4	8.1	6.6	6.9	7.6
	PrecMin	31.3	0.9	0.5	0.1	1.2	0.6	1.4	1.5	0.5	0.1	0.0	0.2	1.5
	PrecStd	7.2	1.3	1.4	1.8	1.1	1.5	1.3	1.6	2.4	2.1	2.0	1.9	1.8
	PrecStd	7.2	1.3	1.4	1.8	1.1	1.5	1.3	1.6	2.4	2.1	2.0	1.9	1.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	55.4	52.2	55.2	62.4	71.2	78.9	81.5	81.8	79.3	74.6	65.9	60.7	
	0.4% MCWB	54.5	50.9	54.3	60.6	70.0	75.8	76.1	77.8	76.3	72.1	64.6	60.0	
	2% DB	53.0	49.9	52.7	59.1	68.8	76.5	80.1	80.2	77.6	71.9	63.9	58.9	
	2% MCWB	52.4	48.6	52.0	58.1	67.5	74.5	76.0	76.4	75.0	69.8	62.6	58.0	
	5% DB	50.8	48.0	50.7	56.7	66.5	75.1	78.9	79.3	76.3	70.2	62.3	57.2	
	5% MCWB	49.7	46.7	49.8	55.3	65.3	73.5	75.7	75.6	73.2	67.6	60.5	56.2	
	10% DB	48.8	46.2	49.0	55.1	64.6	73.8	78.0	78.2	75.2	68.8	61.0	55.3	
	10% MCWB	47.3	44.7	47.6	53.4	63.4	72.2	75.2	74.6	71.5	65.7	58.8	54.0	
	0.4% WB	55.3	52.1	53.3	61.6	69.1	77.4	79.1	80.7	78.1	73.1	65.3	60.7	
	0.4% MCDB	55.6	52.4	53.6	62.9	69.3	78.1	79.4	81.2	78.5	74.1	65.5	60.9	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	52.8	49.7	51.1	58.8	67.0	75.7	78.0	78.3	77.0	71.3	63.5	58.7	
	2% MCDB	53.3	50.1	51.4	59.3	67.3	76.4	78.4	79.2	77.7	72.4	64.1	58.9	
	5% WB	49.9	47.3	49.2	55.9	65.5	74.5	77.2	77.4	75.1	69.5	61.5	56.7	
	5% MCDB	50.4	47.8	49.7	56.4	66.1	75.1	77.8	78.6	76.5	70.7	62.3	57.1	
	10% WB	47.4	45.2	47.2	53.6	64.0	73.2	76.3	76.4	73.3	67.6	59.9	54.3	
	10% MCDB	48.2	45.9	47.9	54.2	64.5	73.8	77.1	78.0	74.7	68.9	60.8	55.1	
(35) Mean Daily Temperature Range	MDBR	7.7	7.2	6.1	5.4	4.7	4.2	4.1	3.7	3.8	5.2	6.1	7.1	
	5% DB MCDBR	8.8	8.2	7.0	6.5	5.8	5.5	4.6	4.5	3.8	5.4	6.3	7.5	
	5% DB MCWBR	10.4	9.7	7.9	5.9	5.4	5.2	4.7	4.8	5.1	7.1	7.9	10.0	
	5% WB MCDBR	8.4	8.4	7.2	6.9	5.0	4.9	3.7	4.0	3.5	4.8	5.6	7.8	
	5% WB MCWBR	10.3	9.9	8.1	6.4	5.2	5.1	4.0	4.4	4.8	6.8	7.6	10.1	
	5% WB MCWBR	10.3	9.9	8.1	6.4	5.2	5.1	4.0	4.4	4.8	6.8	7.6	10.1	
(40) Clear-Sky Solar Irradiance	taub	0.322	0.327	0.357	0.393	0.440	0.484	0.500	0.498	0.413	0.366	0.334	0.322	
	taud	2.512	2.496	2.443	2.357	2.261	2.184	2.152	2.146	2.371	2.497	2.533	2.541	
	Ebn at Noon	272	286	286	281	268	256	250	247	265	268	266	264	
	Edh at Noon	25	29	33	38	43	46	48	47	35	28	24	23	
(44) All-Sky Solar Radiation	RadAvg	659	927	1284	1645	1866	2037	1987	1780	1453	1082	758	579	
	RadStd	60	93	106	120	200	167	104	127	139	102	71	55	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (46 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+250	+142

Nomenclature: See separate page